

Lesson 2 Logical Connectives

In Lesson 1, we introduced the process of proving a theorem using *definitions* and *logic*. In this lesson, we will focus on understanding logical **connectives** such as **not**, **and**, **or**, **if...then**, and **if and only if**. Their meaning in everyday usage can be ambiguous; our goal is to know their *precise* meaning in logic and math so that we can understand math statements correctly.

The connective “**not**” operates on a statement p to produce the **negation** of p , “not p ”, denoted $\sim p$; it is false when p is true and vice versa, as summarized or defined by the following **truth table**:

p	$\sim p$
T	F
F	T

- Let p be “ x is an even integer”, its negation is $\sim p$: “ x is not an even integer”.
- The negation of “ $a > b$ ” is “ $a \not> b$ ”, or equivalently, $a \leq b$.

The connectives “**and**” and “**or**” are *binary* operations (two inputs) which gives:

- conjunction** “ p and q ”, denoted $p \wedge q$, true only when p and q are true.
- disjunction** “ p or q ”, denoted $p \vee q$, false only when both p and q are false.

p	q	$p \wedge q$	$p \vee q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

In logic, “or” is **inclusive**: “ p or q ” means “ p or q or *both*”. In everyday use, “or” may be exclusive or inclusive: “I can have pizza or Thai for lunch” probably does not mean both, but “you get an A if your average is 94 or you ace the final” does not exclude both. Note that “**but**” means “and” in logic (with different nuances).

- Let p be “ x is a positive integer” and q be “ x is an even integer”, then $p \wedge q$ is “ x is a positive even integer”, true when $x = 2, 4, 6, 8, 10, 12, \dots$, while $p \vee q$ is true when x is a positive or even integer (or both): $\dots, -4, -2, 0, 1, 2, \dots$
- The statement “ $2 > 1$ or $2 > 4$ ” is true, while “ $2 > 1$ and $2 > 4$ ” is false.

“**If...then**” combines a **premise/condition/hypothesis** p and a **conclusion** q to produce a **conditional statement** “if p then q ”, or equivalently “ p implies q ”, denoted $p \rightarrow q$, which is false *only* when p is true and q is false.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Pay attention to the third row, $F \rightarrow T = T$! If the premise p is false, then the conditional $p \rightarrow q$ is true, whether q is true or false. Suppose I say, “If it does not rain, then I go jogging.” If it does not rain and I do not go jogging, then the conditional is false, that is the second row: $T \rightarrow F = F$, we all agree. What we find challenging is the third row $F \rightarrow T = T$: if it rains but I go jogging, my statement is also true! The confusion is due to our casual everyday usage. When we say that “If it does not rain, then I go jogging”, we often *also* mean the **inverse** $\sim p \rightarrow \sim q$: “If it rains, then I do not go jogging”, as well as the **converse** $q \rightarrow p$: “If I go jogging, then it does not rain.” In logic, $p \rightarrow q$ is *not* equivalent to its **inverse** or **converse**: $p \rightarrow q$ has a truth table different than $\sim p \rightarrow \sim q$ or $q \rightarrow p$:

p	q	$p \rightarrow q$	$q \rightarrow p$	$\sim p$	$\sim q$	$\sim p \rightarrow \sim q$
T	T	T	T	F	F	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

The conditional “if $x > 2$, then $x^2 > 4$ ” is true, but the converse “if $x^2 > 4$, then $x > 2$ ” is false because $(-3)^2 = 9 > 4$ but $-3 \not> 2$. The Inverse “if $x \leq 2$, then $x^2 \leq 4$ ” is also false because $-3 \leq 2$ but $(-3)^2 = 9 \not\leq 4$.

In logic, if we say “if p then q ”, we mean just that and *not* also “if q then p ” or “if not p then not q ”. If we mean “if q then p ” *and* its converse or inverse (note that they have the same truth table), then we use “**if and only if**”.

- The **biconditional** “ p if and only if q ”, denoted p iff q or $p \leftrightarrow q$, is defined as:

$$p \leftrightarrow q = (p \rightarrow q) \wedge (q \rightarrow p)$$

p	q	$p \rightarrow q$	$q \rightarrow p$	$p \leftrightarrow q = (p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Here is an important and potentially confusing fact, so pay extra attention!

- The conditional “ p if q ” means “if q then p ”, but **in a definition**, “ **p if q** ” means “ **p if and only if q** ”. Some authors explicitly use “if and only if” in a definition but a lot of people just use “if”. **Definitions are biconditionals!**

The definition “An integer x is **even** if $x = 2n$ for some integer n ” means: “An integer x is even *if and only if* $x = 2n$ for some integer n ”, i.e. if x is even, then $x = 2n$ for some integer n , **AND** if $x = 2n$ for some integer n , then x is even.

Finally, we make a connection between logic and arithmetic. If we think of T and F as numbers 1 and 0, then “or” and “and” are almost analogous to $+$ and $\times$, but the analogy is not exact: $T \vee T = T$ but $1 + 1 \neq 1$.